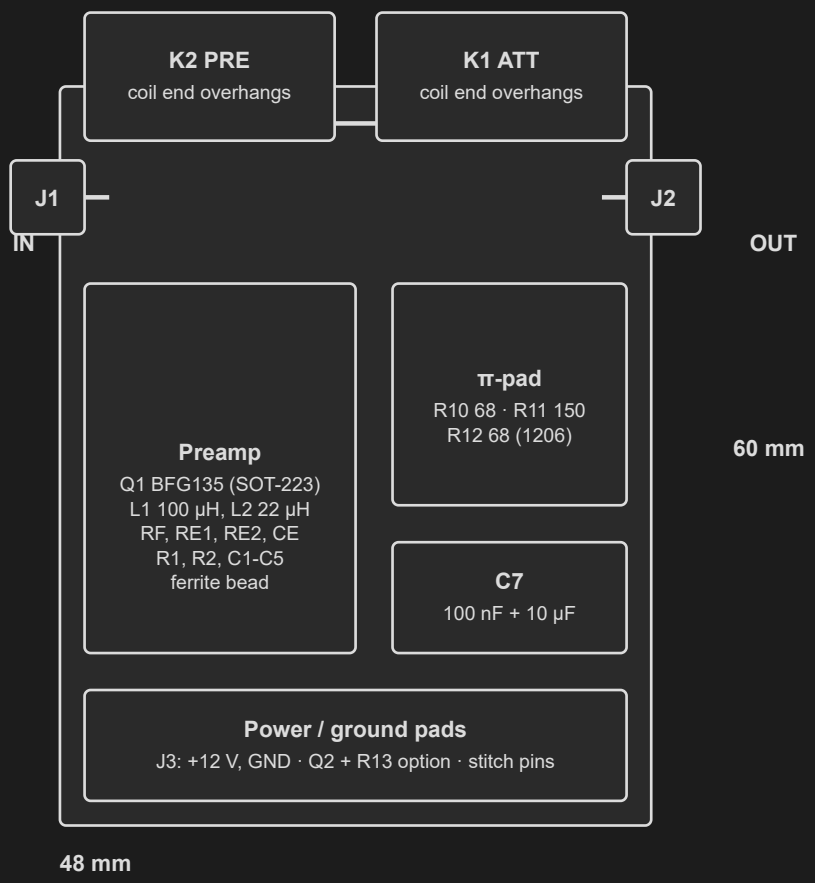
# ATT / PRE board — PCB plan for EasyEDA

48 × 60 mm, on the crystal filter can. IN top left → K2 PRE → K1 ATT → OUT top right. Preamp on the left (away from the Nano), pad on the right.

**Floor plan, to scale (6 px = 1 mm), component side**




## Net list

| Net | Connects |
|---|---|
| RF_IN | J1 signal, K2 COM-A |
| PRE_BYP | K2 NC-A, K2 NC-B (short wide track) |
| PRE_IN | K2 NO-A, C1 |
| BASE | C1, Q1 B, L2, RF-feedback (RF), (bias via L2) |
| BIAS | L2, R1, R2, C5 |
| EMIT | Q1 E, RE1 |
| E2 | RE1, RE2, CE |
| COLL | Q1 C (tab), ferrite bead |
| COLL2 | ferrite bead, L1, C4, C3 |
| FB | C3, RF |
| PRE_OUT | C4, K2 NO-B |
| MID | K2 COM-B, K1 COM-A |
| ATT_BYP | K1 NC-A, K1 NC-B (short wide track) |
| PAD_IN | K1 NO-A, R10, R11 |
| PAD_OUT | R11, R12, K1 NO-B |
| RF_OUT | K1 COM-B, J2 signal |
| V12 | J3-1, C7, R6 (Q2 emitter if fitted) |
| V12F | R6, C6, R1, L1 (Q2 collector feeds R6 if fitted) |
| GND | J1/J2 shields, J3-2, R2, R10, R12, RE2, CE, C5, C6, C7, bottom plane |

Relay pin names are generic (COM / NC / NO, poles A and B): map them to the Finder 30.22 footprint (NC = rest position). K2 = PRE, K1 = ATT. Values on the ATT / PRE board sheet.

## Design rules (home etching)

- Parts 1206 (resistors, capacitors), BFG135 SOT-223, chokes 1210 or axial.
- RF tracks **1.5 mm**, power **2 mm**, others ≥ 1 mm; clearance ≥ **0.6 mm**.
- Top: tracks + ground pour (0.8 mm clearance). Bottom: **unetched solid copper = ground plane**.
- Top ground to bottom: wire pins through drilled holes, at least next to every GND pad of the RF parts, at J1 / J2 and around the preamp.
- Relay holes: no relay pin is GND — clear the bottom copper (ø 2.5 mm) around **every** relay pin and every other non-GND through-hole.

- Coil pins overhang the edge: flyback 1N4148 and sense diode soldered directly on the coil pins, wires to the rocker / Nano.